



Diabetes Prediction Using Multilayer Perceptron (MLP)

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Introduction

Diabetes is one of the most prevalent chronic diseases in the world and early diagnosis is important to improve patient health. Machine learning techniques can be used by medical professionals to analyze patient data and predict whether or not a person is likely to have diabetes.

A Multilayer Perceptron (MLP) neural network has been implemented in this project to predict whether the patient is diabetic or non-diabetic using medical diagnostic measurements. The project revolved around the pre-processing of the dataset, designing and training the neural network, evaluating its performance and running multiple experiments with varying hyperparameters.

Method

The implemented solution uses a Multilayer Perceptron (MLP), which is a type of Artificial Neural Network designed for classification tasks.

The project workflow looked as follows:

1. Data Standardization

- Load the dataset.
- Checking for missing or incorrect values.
- Feature scaling for normalizing the numeric features.
- Divide data into training , validation and testing sets.

2. Models Structure

- An input layer with the same number of features as the data set.
- Two hidden layers with activation function ReLU.
- Output layer with sigmoid activation function for binary classification.

3. Training Process

- The model trained by backpropagation and gradient descent optimization.
- Binary Cross Entropy loss was the loss function used.
- Training and validation loss values were monitored.

4 Evaluation

- The trained model was tested on the testing dataset.
- The performance metrics such as accuracy and final loss were calculated.

Dataset Description

The project was based on Diabetes Dataset, a dataset of medical information of patients and it includes 700 rows .

The aim of the dataset is to predict if the patient is diabetic or not. The dataset contains a number of medical attributes, such as:

- ✓ Pregnancies
- ✓ Glucose
- ✓ Blood pressure
- ✓ Skin thickness
- ✓ Insulin
- ✓ Body mass index (BMI)
- ✓ Age

The dependent variable is:

0 → No diabetes

1 → Diabetes

The dataset was preprocessed to deal with invalid values by using median to save data as much as can and scale the numerical features before training the model to improve model performance and training stability.

You can visit dataset from:

<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>

Experiment

A series of experiments were performed to evaluate the impact of hyperparameters on the effectiveness of models with different learning rates. The model developed was a Multilayer Perceptron (MLP) for binary classification.

The final architecture of the neural network was:

- input layer 8 features
- First hidden layer (32 neurons)
- Second hidden layer, 16 neurons
- Batch Normalization was applied after each hidden layer to stabilize training .
- Activation function ReLU was used in hidden layers
- Dropout was used with a value of 0.3 to reduce overfitting
- Sigmoid activation function in the output layer

The optimizer and loss function used were the following:

- Adam optimizer and Binary Cross-Entropy Loss (BCELoss)
- Learning rate 0.0025
- Step Learning Rate Schedule

The scheduler used to reduce the learning rate by a factor of 2 every 30 epochs for the stability of training.

Hyperparameters play an important role in neural network performance. Several hyperparameters directly affected the behavior of the model during training.

Results

1)The learning rate controls how much the model weights are updated during optimization.

- 0.0025 produced stable learning and better accuracy.
- 0.001 caused instability and reduced accuracy.
- 0.005 caused instability and reduced accuracy to 30%.

Experiment 1	Experiment 2
I. Learning rate = 0.0025 II. Dropout=0.3 III. The accuracy of the model on the testing data set was around 72%	I. The learning rate was increased from 0.0025 to 0.005 and to 0.001 II. All other parameters remained the same III. The accuracy of the model on the testing data set was around 30%

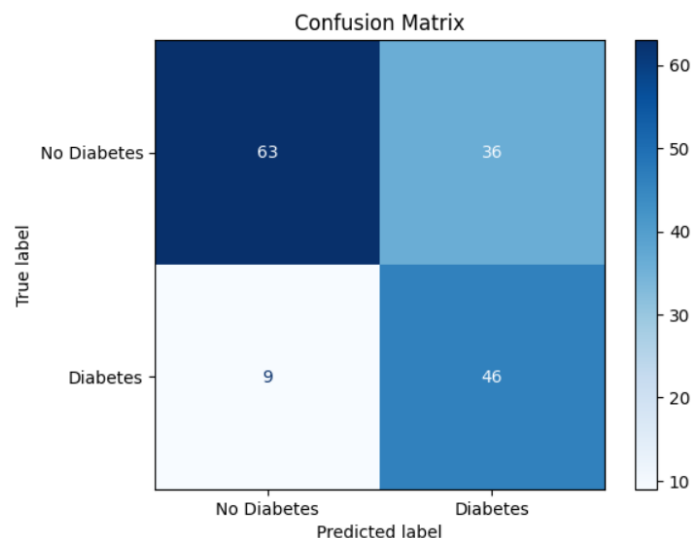
Confusion Matrix

The model correctly classified:

- **63** non-diabetic patients as *No Diabetes*.
- **46** diabetic patients as *Diabetes*.

However, there were some misclassifications:

- **36** non-diabetic patients were incorrectly predicted as diabetic (False Positives).
- **9** diabetic patients were incorrectly predicted as non-diabetic (False Negatives).



The relatively small number of false negatives is important in medical diagnosis because missing a diabetic patient may lead to serious health risks. Although the model produced some false positives, it was generally more successful

Discussion

The results show that neural networks can be a good solution to medical diagnosis classification problems. Adding hidden layers and neurons increased the model's ability to learn complex relationships in the data.

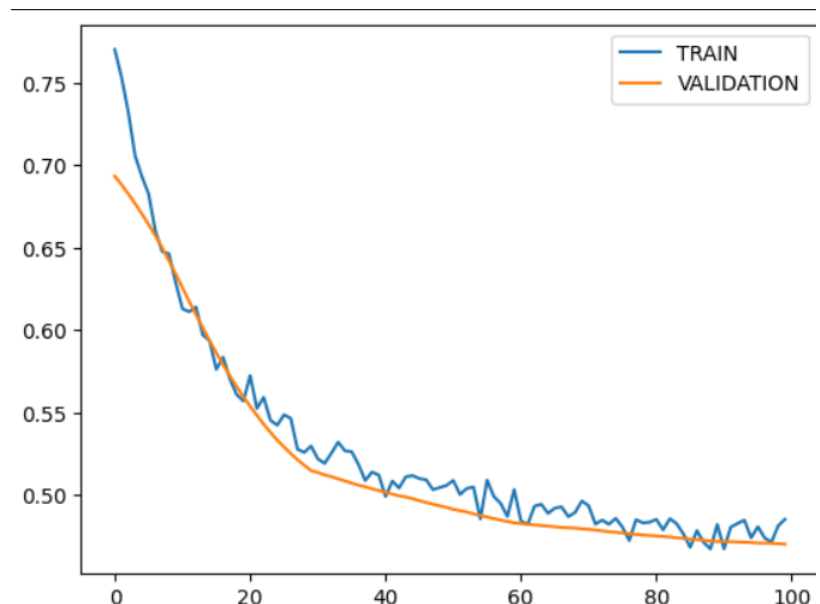
However, several challenges were seen:

- The size of the dataset was relatively small (around 700 rows).
- Some features contained either zero or invalid values.
- If the network is too complex it may become overfitted.
- How to choose the optimal learning rate.
- Training was stabilized and improved by normalization and appropriate learning rates.

Loss Curves

The loss curve reveals that the training loss and validation loss have dropped gradually during the training process, which means the model has learned from the dataset successfully.

The two curves are close to each other which is indicative of good generalization and small overfitting. And the loss values stabilized for the last epochs, which means the model has converged well.



Conclusion

This project implemented a diabetes prediction system using Multilayer Perception neural network. The model could classify diabetic and non-diabetic patients based on the medical measurements.

The experiments proved that the neural network parameters such as the number of hidden layers and the learning rate can greatly affect the model performance. Overall the project proved the practical use of neural networks in healthcare and medical diagnosis systems.

Future Work

We can do a number of improvements for future versions of the project:

- *Leveraging larger and more diverse medical datasets.
- *Leveraging advanced deep learning models.
- *Doing hyperparameter optimization automatically.
- *Deploying the model as an application for real time predictions.